**EXPECTED AI LANDSCAPE**

| AI Categories | Tools commonly used | Data flowing through tools and imputed by engineers | Indicator questions |
|---|---|---|---|
| Development AI | • GitHub Copilot (Individual, Business, Enterprise)<br>• Cursor<br>• Claude Code, Codeium, Continue<br>• ChatGPT, Claude, Gemini for engineering | • Code context and prompts.<br>• User engagement data (interactions like accepts or dismissals).<br>• Telemetry data such as interaction logs and types of queries submitted.<br>• Browser and UI verification data.<br>• Model context protocol and third-party data.<br>• Data and rich content files.<br>• Language server protocol data.<br>• Natural language inputs such as text prompts. | • When codes and software architecture are generated with these tools, where does the source content come from?<br>• Who in your company reviews the codes and software material?<br>• How technically trained are the personnel in charge of this? |
| Engineering-specific AI | • Computer-Aided Design (CAD) and Electronic Design Automation (EDA) AI<br>• Requirements management AI<br>• Test and verification AI<br>• Model-based design AI<br>• Printed Circuit Board (PCB) and layout AI | • Textual specs and metadata including technical specifications, engineering handbooks, and natural language prompts.<br>• Business documentation and market requirement documents.<br>• Interface guidelines including wireframes and UX/UI design assets.<br>• Manufacturing tolerances including minimum trace widths and minimum spacing between copper elements. | • What is your company's policy on AI assistance in safety-critical code versus non-safety-critical code?<br>• How periodically do you perform a maintenance of your system(s)?<br>• How do you check for anomalies in performance of your system(s)? |

| | | | |
|---|---|---|---|
| Operations and Productivity AI | • Microsoft 365 Copilot<br>• Google Workspace AI<br>• Proposal automation AI<br>• Meeting AI | • Uploaded reference files such as a Word document, PDF or CSV.<br>• Communication content such as emails, Outlook attachments, and instant messages from Team chats.<br>• Media repositories such as audio/video files stored in Google Drive.<br>• Calendar and invite specifics such as meeting titles, attendee lists, organisation charts, scheduled durations, and baseline meeting agenda or description. | • Are you aware of the presence of an audit trail when using these tools?<br>• How secure is your company's information while using these tools and are your personnel sensitised regarding this?<br>• How frequently do you utilise these tools and in what types of engagements? |
| AI in product features | • Automotive: Advanced Driver Assistance Systems<br>• Industrial IoT (Internet of Things): Predictive Maintenance Sensors<br>• Medical Devices: Smart Wearables and Diagnostics | • Sensor calibration matrices.<br>• FlexRay data which are time-triggered, deterministic data loops handling highly safety-critical information like electronic power steering and active braking feedback.<br>• High-frequency raw acceleration data, and frequency-domain data.<br>• Ultrasonic and acoustic emission data.<br>• High-speed pressure data capturing fluid compression cycles.<br>• Contextual and operational metadata such as external atmospheric temperature and humidity.<br>• Physiological and biometric data.<br>• Device calibration and impedance metrics. | • How do you check for regulatory compliance of your system(s)?<br>• Are you building AI features into products you ship? How are those features validated?<br>• Explain how your company receives and measures feedback from clients regarding the quality of your product features? |

**Key discovery probing questions**

1. When codes and software architecture are generated, what automated tools or systems are used and where does the source content come from?

2. How secure is your company's information while using these tools and are your personnel sensitised regarding this?

3. How do you check for regulatory compliance of your products and services, and what regulatory frameworks are they aligned with?

4. Are you building features into your products that may possibly have AI components? How are those features validated?

5. Explain how your company receives and measures feedback from clients regarding the quality of your product features?

**Expected sticking points or pushback (in response to the aforementioned questions), along with constructive responses in *italics*.**

1. The source content derives mainly from our company files or records but we can check again and provide more specific details on this.

   *Okay, ensure to consider sources including your company's proprietary hardware and software data, firmware source code, records of design specifications and operational workflows, among other sources.*

2. All our staff have an awareness of what kind of company information is shared.

   *Well understood but it's vital to be aware of the specific types of company data entered into various systems, and if they happen to have AI components, we can provide guidance on how such data is utilised and the kind of data-handling protocols that can help.*

3. Our products comply with regulatory laws which were stipulated at the start of our company many years ago.

   *That's great. Let's go through each of your main products and services. This way we can more easily ascertain if there are some that fall within the governance of AI regulations. In such cases, we would need to discuss more about the NIST AI Risk Management Framework, and at the state-level, the Texas Responsible Artificial Intelligence Governance Act.*

4. We mainly use AI tools such as Claude or Gemini to aid documentation but need to doublecheck if and what type of AI features are actually embedded in our products.

   *Indeed, the inventory you provided also does not indicate specific AI tools. However, beyond your use of Claude and Gemini, our AI landscape document can provide a thorough guide as you doublecheck your systems and products.*

5. We normally receive feedback from clients via our website or verbally but we will look into a more systematic method of obtaining this.

   *Okay, we can examine your current feedback retrieval methods to check if any form of AI system is involved. Additionally, we can propose a number of ways through which you can receive and measure feedback on your products and services through the use of automated device metrics.*